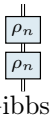
Scope of the Spacetime Bootstrap

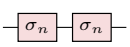
(Dated: October 15, 2026)

The spacetime bootstrap establishes a uniform spectral gap from bond reflection positivity and certified finite-scale purity bounds. Bond reflection positivity ensures that the continuous-time spatial transfer operator is self-adjoint, enabling spatial scale doubling dual to imaginary time. We formalize the method as a general criterion for one-dimensional translation-invariant lattice systems, extend it to multi-site unit cells with internal reflection and odd seed lengths, classify standard spin and ladder Hamiltonians, and summarize the computer-verified gap certificates across eight model families.

Appendix D: Prerequisites of the Spacetime Bootstrap

The scale-doubling iteration in Ref. [1] comprises three elementary operations:

- *Coupling identities.* Equalities between partition-function ratios on the spacetime torus, such as $\square/\square^4$. These hold universally for every translation-invariant lattice Hamiltonian.
- *Temporal squaring.* The relation 

$\propto \rho_n(2\beta)$ requires only positivity of the Gibbs operator, $e^{-\beta H} = e^{-\beta H/2}(e^{-\beta H/2})^\dagger \geq 0$, which holds generally by Euclidean time reflection.
- *Spatial squaring.* The dual relation 

$\propto \sigma_{2n}$ requires positivity of the spatial transfer operator, $\sigma_n = X_\beta^n/Z_n \geq 0$. This holds whenever X_β is self-adjoint ($X_\beta = X_\beta^\dagger$), which additionally guarantees that $Z_n = \sum_i \lambda_i^n$ with real eigenvalues λ_i, permitting ℓ^p norm interpolation to non-dyadic intermediate lengths.

Self-adjointness of X_β represents the spatial counterpart of Euclidean time reflection, and is not automatic. Once it holds, the remaining doubling argument follows directly. The three hypotheses are therefore:

(H1) *Locality.* A translation-invariant chain with finite local dimension and nearest-neighbor terms. Finite-range interactions reduce to this by blocking sites into unit cells.

(H2) *Bond reflection positivity* (Sec. E).

(H3) *Input.* Both purities are near one at a single scale (Sec. F).

Hypotheses (H1) and (H2) are algebraic properties of the Hamiltonian, whereas (H3) reflects the underlying spectral gap.

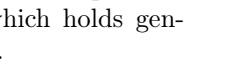
Appendix E: Reflection positivity

Let Θ be an antiunitary map on one site with $\Theta^2 = \pm 1$, for example complex conjugation K in a chosen basis. For an operator C on one site write $\Theta(C) = \Theta C \Theta^{-1}$.

- *Reflection.* Cut the chain between two neighboring sites. Mirror the chain across the cut and apply Θ on every site. Call the result θ. It is antilinear, it maps the operators on the left half to those on the right half, and $\theta^2 = 1$.
- *Bond form.* A nearest-neighbor Hamiltonian is in *bond reflection-positive form* [6–9] if it reads

$$H = \sum_j D_j - \sum_{j,k} c_k \, C_{k,j} \, \Theta(C_k)_{j+1}, \qquad c_k > 0, \quad \text{(E1)}$$

with $\Theta(D) = D$. Each bond event of weight c_k inserts C_k on one site and its mirror image $\Theta(C_k)$ on the next site,

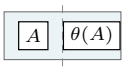
$$-h_{j,j+1} = \sum_k c_k$$ 

.

The sign in front matters. A term with $c_k < 0$ cannot be absorbed by rescaling C_k, because Θ conjugates the factor i: $(iC) \otimes \Theta(iC) = C \otimes \Theta(C)$.

The model of Ref. [1] has this form with $\Theta = K$ in the Cartesian basis and $c_k = 1$. Its generators $G_\alpha = iS^\alpha$ are real, so $-\boldsymbol{S} \cdot \boldsymbol{S} = \sum_\alpha G_\alpha \otimes \overline{G_\alpha}$.

Reflection positivity and transfer matrices. Expand $e^{-\beta H}$ in the bond events that cross the mirror. Each term is a left half followed by its mirror image, so for every operator A on the left half,

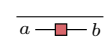
$$\mathrm{Tr}\big[A\,\theta(A)\,e^{-\beta H}\big] =$$ 

$= \sum \langle \Phi_A | \Phi_A \rangle \geq 0.$ (E2)

Reflection positivity began as the Osterwalder–Schrader axiom that rebuilds a Hilbert space and a positive transfer matrix from Euclidean data [3]. On the lattice it gives a self-adjoint, positive transfer matrix [4], and its Cauchy–Schwarz structure drives infrared bounds and chessboard estimates [5, 7, 8]. For spins and Majoranas, Jaffe and Janssens characterize exactly which couplings are reflection positive [6]. The spacetime bootstrap needs only the weaker consequence that follows: the network cut at a single bond is Hermitian. This is the Lüscher statement with space and time exchanged.

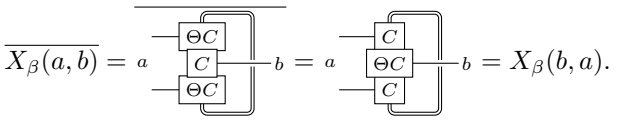
Lemma 1 (Mirror is conjugation)**.** *Under Eq.* (E1)*, build the continuous-time MPO as in Ref. [1]. The events*

are (τ, k), they occur at rate c_k, and $e^{-\tau D}$ propagates between them. Then the transfer kernel obeys



$$= \quad \text{(same diagram, } a \leftrightarrow b\text{)}, \qquad \textit{i.e.} \quad \overline{X_\beta(a,b)} = X_\beta(b,a). \tag{E3}$$

Proof. Open the vertex in imaginary time. Events on the left bond a insert $\Theta(C)$, and events on the right bond b insert C, along a single traced physical loop:



Write $\Theta = VK$. Complex conjugation of a trace conjugates every insertion and keeps the time order. Conjugating the loop by V then maps $\overline{\Theta(C)} \mapsto C$, $\overline{C} \mapsto \Theta(C)$ and $\overline{D} \mapsto D$, because $V\overline{V} = \pm 1$ and $\Theta(D) = D$. The picture that results is the mirror image of the original: the left insertions are now C and the right ones are $\Theta(C)$. □

Theorem 1 (Space behaves like time)**.** *Assume (H1) and (H2). Then X_β is a self-adjoint Hilbert–Schmidt operator on $L^2(\mu_\beta)$. For every even n, $Z_n(\beta) = \sum_i \lambda_i(\beta)^n$ with real λ_i, and $\sigma_n(\beta) \geq 0$. In particular the squaring lemma, the coupling identities and the ℓ^p interpolation of Ref. [1] hold verbatim.*

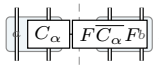
Proof. The event measure μ_β is positive because $c_k > 0$, and it is finite, with mass $e^{\beta \sum_k c_k}$ per bond. Normalize $\|C_k\| = 1$. The kernel is then bounded, $|X_\beta(a,b)| \leq d\, e^{\beta \|D\|}$, hence Hilbert–Schmidt, and it is Hermitian by Lemma 1. The remaining statements are those of Ref. [1] for a self-adjoint X_β. □

Multi-site unit cells and internal reflection. When c sites are blocked into a unit cell on a ring of L cells, $H_L = \sum_{j=1}^{L} h_{j,j+1} + \sum_{j=1}^{L} o_j$, a mirror cut across an inter-cell bond swaps the left and right virtual legs of each cell while reversing the site order inside the cell. Let F be a real symmetric involution ($F = F^{\mathsf{T}}$, $F^2 = I$) that reverses the internal site order. If the inter-cell bond h and on-site cell operator o satisfy

$$-h = \sum_\alpha C_\alpha \otimes F\overline{C_\alpha}F, \qquad FoF = o, \qquad o = \overline{o}, \tag{E4}$$

then $\Theta(C) = F\overline{C}F$ is antiunitary with $\Theta^2 = 1$ and $\Theta(o) = o$. Each inter-cell bond event pairs site b of the left cell with its reflected partner a in the right cell,

$$-h_{j,j+1} = \sum_\alpha$$



and conjugating the physical loop by F in Lemma 1 gives $\overline{X_\beta(a,b)} = X_\beta(b,a)$ on the ring of L cells (`RPRing` in `lean/CertifiedGaps/Ring/Model.lean` and `Ring/Transfer.lean`). For a two-spin dimer (a,b) with the two-spin swap F, $C_\alpha = I \otimes G_\alpha$ gives $F\overline{C_\alpha}F = \overline{G_\alpha} \otimes I$, linking spin b of cell j to spin a of cell $j+1$. The internal swap F is essential: any unreflected real form $\sum_\alpha C_\alpha \otimes C_\alpha$ ($F = I$) is symmetric under the two-cell swap $S(x \otimes y) = y \otimes x$, whereas the inter-cell bond has $\|h - ShS\|_{\mathrm{F}}^2 = 6(1-\delta)^2 > 0$ for $\delta \neq 1$ in every single-cell basis (`models/dimerized_heisenberg`). Diagonalizing F orthogonally and multiplying its -1 eigenspace by i reduces $\Theta(C) = F\overline{C}F$ to entrywise conjugation.

Positive cell-transfer operators and odd seed lengths. Theorem 1 restricts $Z_n(\beta) = \sum_i \lambda_i(\beta)^n$ to even n because a self-adjoint X_β may have negative eigenvalues. In a two-spin dimer cell (a,b) whose intra-cell bond is also reflection positive with real factors, cutting the cell across its internal bond factors the cell transfer operator into a half-cell product,

$$X_\beta = \quad$$



$$= KK^* \succeq 0. \tag{E5}$$

Every transfer eigenvalue $\lambda_i(\beta)$ is then nonnegative, so $\sigma_n(\beta) = X_\beta^n / Z_n(\beta) \geq 0$, the squaring map, and ℓ^p interpolation hold at odd as well as even cell counts n_0. In the dimerized spin-1 chain (`models/spin1_dimerized`), Eq. (E5) enables an odd seed of $n_0 = 3$ dimer cells (6 and 12 spins), bypassing the $2n_0 = 8$-cell (16-spin, $3^{16} = 4.3 \times 10^7$-state) ring required by $n_0 = 4$.

Relation to the sign problem. A Hamiltonian is stoquastic if, in some basis, every configuration of a quantum Monte Carlo expansion has a nonnegative weight. Reflection positivity asks less of individual configurations and more of their structure: a weight may be negative, provided each configuration paired with its mirror image contributes $|\cdot|^2$.

- In Ref. [1] the event measure is positive, but the entries of W and X_β carry signs, because the G_α are real antisymmetric matrices. A sampler over histories would meet a sign problem; the spectral argument does not, since $\sum_i \lambda_i^n > 0$.
- Neither property implies the other: the Heisenberg ferromagnet is stoquastic but not reflection positive [10] (Proposition 1).
- Sign-free fermion Monte Carlo methods based on time reversal or Majorana symmetry use the same mirror-squared factorization $\det M \cdot \overline{\det M}$ [11–14].

Appendix F: The general criterion

Define the one-step loss

$$\delta(\varepsilon) = \tfrac{1}{2}\left[(1-\varepsilon)^{-3} - 1\right]^2. \tag{F1}$$

It lies below the diagonal,  , up to $\varepsilon_* \simeq 0.1273$, the root of $\delta(\varepsilon) = \varepsilon$. Ref. [1] uses the cruder bound $\delta \leq 5\varepsilon^2$, valid for $\varepsilon \leq 1/100$.

Theorem 2 (Spacetime-bootstrap criterion)**.** *Assume (H1) and (H2), and suppose that for some even n_0 and some $\beta_0 > 0$,*

$$P_{\mathrm{s}}(n_0, \beta_0), \;\; P_\tau(2n_0, \beta_0) \geq 1 - \varepsilon_0, \qquad \varepsilon_0 < \varepsilon_*. \tag{F2}$$

Set $\varepsilon_{j+1} = \delta(\varepsilon_j)$, $n_j = 2^j n_0$ and $\beta_j = 2^j \beta_0$. Then for every even $L \geq n_0$ the ground state is unique and

$$\Delta_L \;\geq\; c := \inf_{j\geq 0} \frac{1}{\beta_j} \log \frac{1-3\varepsilon_j}{3\varepsilon_j} \;>\; 0. \tag{F3}$$

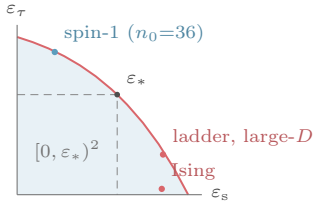
Proof. By Theorem 1, the four moves of the doubling estimate in Ref. [1] apply without change. Move 1 gives $P_{\mathrm{s}}(n, 2\beta) \geq (1-\varepsilon)^3$. Move 2 squares in space to get $P_{\mathrm{s}}(2n, 2\beta) \geq 1 - \delta(\varepsilon)$. Move 3 gives $P_\tau(4n, \beta) \geq (1-\varepsilon)^2(1-\delta) \geq (1-\varepsilon)^3$, which requires $\delta \leq \varepsilon$, i.e. $\varepsilon \leq \varepsilon_*$. Move 4 squares in time to get $P_\tau(4n, 2\beta) \geq 1 - \delta(\varepsilon)$. Induction on j gives $P_{\mathrm{s}}(n_j, \beta_j), P_\tau(2n_j, \beta_j) \geq 1 - \varepsilon_j$, with ε_j decreasing because $\delta(\varepsilon) < \varepsilon$ on $(0, \varepsilon_*)$. Since $3\varepsilon_j < 1/2$, the interpolation and gap-recovery steps of Ref. [1] give uniqueness and $e^{-\beta_j \Delta_L} \leq 3\varepsilon_j/(1-3\varepsilon_j)$ whenever $n_j \leq L \leq 2n_j$. Every term of the infimum is positive; since $\delta(\varepsilon) \leq K\varepsilon^2$ near zero, ε_j falls doubly exponentially, so the terms converge to a positive limit. □

With $(n_0, \beta_0, \varepsilon_0) = (2304, 784, 1/100)$ this is the theorem of Ref. [1], with $c \geq (\log 20)/784$.

Asymmetric defects and half-length seeds. In Theorem 2, the initial spatial defect $\varepsilon_{\mathrm{s}} \geq 1 - P_{\mathrm{s}}(n_0, \beta_0)$ and thermal defect $\varepsilon_\tau \geq 1 - P_\tau(2n_0, \beta_0)$ need not be equal. Tracking $(\varepsilon_{\mathrm{s},j}, \varepsilon_{\tau,j})$ separately through Moves 1–4 gives the asymmetric recurrence

$$\begin{aligned} \varepsilon_{\mathrm{s},j+1} &= f\big(1 - (1-\varepsilon_{\mathrm{s},j})^2(1-\varepsilon_{\tau,j})\big), \\ \varepsilon_{\tau,j+1} &= f\big(1 - (1-\varepsilon_{\tau,j})^2(1-\varepsilon_{\mathrm{s},j+1})\big), \end{aligned} \tag{F4}$$

where $f(z) = z^2/[2(1-z)^2]$. Because the second line uses the updated spatial defect $\varepsilon_{\mathrm{s},j+1}$ from Move 2 rather than $\varepsilon_{\mathrm{s},j}$, the basin of attraction in the $(\varepsilon_{\mathrm{s}}, \varepsilon_\tau)$ plane extends well outside the symmetric square $[0, \varepsilon_*)^2$, reaching $\varepsilon_{\mathrm{s}} \simeq 0.2176$ as $\varepsilon_\tau \to 0$ and $\varepsilon_\tau \simeq 0.2006$ as $\varepsilon_{\mathrm{s}} \to 0$:



Certified seeds in both wings outside $[0, \varepsilon_*)^2$ close under Eq. (F4): the spin-1 seed $(0.0479, 0.1815)$ of Ref. [2] lies in the upper wing, while the $J_\perp = 5/2$ ladder $(0.1856, 0.0511)$, $D = 3.206$ large-D $(0.1916, 0.0465)$, and $(g, h) = (1, 1/20)$ Ising $(0.1840, 0.0074)$ seeds lie in the right-hand wing.

Furthermore, the coupling identity of Ref. [1] expresses $P_\tau(2n_0, \beta_0)$ in terms of length-n_0 purities,

$$P_\tau(2n_0, \beta_0) = \frac{\square}{\square^2} = \frac{P_{\mathrm{s}}(n_0, 2\beta_0)\, P_\tau(n_0, \beta_0)^2}{P_{\mathrm{s}}(n_0, \beta_0)^2}, \tag{F5}$$

bounding $P_\tau(2n_0, \beta_0)$ from length-n_0 and length-$2n_0$ partition functions at β_0 and $2\beta_0$ without a ring of length $4n_0$. At $D = 4$ (`models/spin1_largeD`), combining Eq. (F5) at $2n = 12$ with transfer squaring lifts the seed from $n_0 = 6$ to $n_0 = 12$ using only 6- and 12-site spectra, raising the gap bound from 0.695 to 0.899.

Corollary 1 (What (H3) excludes)**.** *Under (H1) and (H2), Eq. (F2) cannot hold at any scale*

(i) if the ground state is k-fold degenerate at the length $2n_0$ with $k \geq 2$, because $P_\tau \leq w_0 \leq 1/k$ there, a spectrum like ▮▮_;

(ii) if $\liminf_L \Delta_L = 0$ along even L, because Theorem 2 would contradict it.

Corollary 1 rules out gapless, symmetry-broken, and asymptotically degenerate chains on the ring.

The converse is expected but not proved. In a gapped chain with a unique ground state and correlation length ξ, one expects $1 - P_{\mathrm{s}} \sim e^{-n/\xi}$ and $1 - P_\tau \sim n\, e^{-\beta\Delta}$. Both vanish along the doubling sequence ▫▫▭, so Eq. (F2) should hold at some finite scale.

For a conformal critical chain the picture is different. The extensive free energy cancels in each torus ratio, so P_{s} and P_τ approach ratios of the universal torus partition function at a fixed aspect ratio. These are numbers below one, so the defect stalls instead of collapsing.

Appendix G: Classification of Reflection-Positive Hamiltonians

Bilinear spin couplings. Consider spin S with a nearest-neighbor coupling matrix J (real symmetric) and a uniform field $\boldsymbol{h}$,

$$H = \sum_j \boldsymbol{S}_j^{\mathsf{T}} J\, \boldsymbol{S}_{j+1} + \boldsymbol{h} \cdot \boldsymbol{S}_j. \tag{G1}$$

Proposition 1. *Suppose there is a symmetric orthogonal O with $\det O = -1$ such that $-JO \succeq 0$ and $O\boldsymbol{h} = \boldsymbol{h}$. Then Eq. (G1) has the form of Eq. (E1) for every S. For diagonal $J = \mathrm{diag}(J_x, J_y, J_z)$ with nonzero entries and $\boldsymbol{h} = 0$, such an O exists if and only if an even number of J_x, J_y, J_z are negative.*

Proof. $O = -R$ with R the identity or a π rotation. Let T be time reversal, so $TST^{-1} = -S$, and put $\Theta = U_R T$. Then $\Theta \boldsymbol{S} \Theta^{-1} = O\boldsymbol{S}$. Since U_R commutes with T, $\Theta^2 = U_R^2 T^2 = \pm 1$. Factor $-JO = \sum_k v_k v_k^{\mathsf{T}}$ with real v_k and set $C_k = v_k \cdot \boldsymbol{S}$. Then $\sum_k C_k \otimes \Theta(C_k) = \sum \boldsymbol{S}^{\mathsf{T}}(-JO)O\boldsymbol{S} = -\boldsymbol{S}^{\mathsf{T}} J \boldsymbol{S}$, and $\Theta(\boldsymbol{h} \cdot \boldsymbol{S}) = (O\boldsymbol{h}) \cdot \boldsymbol{S}$. For diagonal J, $O = \mathrm{sign}(-J)$, so $\det O = -1$ counts the ferromagnetic entries modulo two. □

The proposition sorts the standard spin chains as follows.

- *Reflection positive:*
  - the Heisenberg antiferromagnet for any S ($O = -1$);
  - XXZ chains $S^x S^x + S^y S^y + \Delta S^z S^z$ with $\Delta \geq 0$;

  - the XY model with any signs ($J_z = 0$ frees O_{zz});
  - the transverse-field Ising chain with either sign (O fixes the field axis), including a longitudinal field $-h\sum_j Z_j$ ($C_1 = Z$, $o = -gX - hZ$).

- *Not of this form:*
  - the Heisenberg ferromagnet (three negative entries);
  - XXZ chains with $\Delta < 0$;
  - the antiferromagnet in a uniform field, since $O = -1$ reverses $\boldsymbol{h}$.

Sublattice rotations by π change two signs at once and never alter the bond parity, but they reverse a *staggered* field $-h\sum_j(-1)^j S^z_j$, ↑↓↑↓: applying $U = 2iS^y$ on odd sites maps the antiferromagnetic bond to $-h_{\mathrm{rot}} = X \otimes X + G_y \otimes G_y + Z \otimes Z$ ($G_y = iS^y$) while negating S^z on odd sites, converting the staggered field into a uniform reflection-even term $+h\sum_j S^z_j$ with a single-spin cell (`models/heisenberg_staggered_field`). Similarly, in Ref. [1] (Appendix A) the open chain is built with opposite endpoint fields ↑•••••↓ because $\theta(-hS^z_1) = +hS^z_n$ is mirror symmetric, and the equal-field chain ↑•••••↑ is reached via U_P.

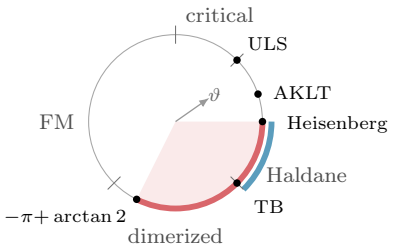
Spin-1 bilinear–biquadratic chain. Consider $H = \sum_j \cos\vartheta\, \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1} + \sin\vartheta\, (\boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1})^2$. In the Cartesian basis the quadrupoles $Q_{\alpha\beta} = S^\alpha S^\beta$ are real. Their antisymmetric part is $\frac{1}{2}\epsilon_{\alpha\beta\gamma}G_\gamma$, so

$$-h = \left(\cos\vartheta - \tfrac{1}{2}\sin\vartheta\right)\sum_\alpha G_\alpha \otimes \overline{G_\alpha} - \sin\vartheta \sum_{\alpha,\beta} Q^+_{\alpha\beta} \otimes \overline{Q^+_{\alpha\beta}}, \tag{G2}$$

where $Q^+_{\alpha\beta} = \frac{1}{2}(Q_{\alpha\beta} + Q_{\beta\alpha})$; the cross terms cancel between symmetric and antisymmetric parts.

Proposition 2. *With $\Theta = K$ in the Cartesian basis, the bilinear–biquadratic chain is in reflection-positive form if and only if $-\pi + \arctan 2 \le \vartheta \le 0$.*

Proof. The condition holds if both coefficients in Eq. (G2) are nonnegative, which is the stated window. Conversely, the form exists exactly when the realigned bond matrix is positive semidefinite after an on-site shift; maximizing its least eigenvalue over all shifts gives a negative value outside the window (-0.087 at $\vartheta = -0.66\pi$ and -0.156 at $\vartheta = 0.05\pi$; Sec. H). □



The red arc is the reflection-positive window; the blue arc, $-\pi/4 < \vartheta \le 0$, is where it meets the Haldane phase so Theorem 2 applies. Within this arc, scans up to $n_0 = 6$ (`models/blbq_haldane`) show a positive ground-energy doubling mismatch $E_0(2n) - 2E_0(n) \in [0.365, 0.920]$ at $n = 6$, which caps $P_{\mathrm{s}}(n,\beta) \lesssim e^{-\beta[E_0(2n)-2E_0(n)]}$ against $\beta \gtrsim 1/\Delta$ and delays admissible seeds to $n_0 \approx 20$–48. Elsewhere in the window Corollary 1 excludes it (degenerate dimerized ground states and the critical Takhtajan–Babujian point TB). The Heisenberg chain sits on the edge of the window, while AKLT and ULS lie outside it because the positive biquadratic coefficient has the wrong sign, and optimizing over all on-site Θ finds none (Sec. H).

Anisotropy, dimerized cells, and ladders. Any real symmetric on-site or intra-cell operator $o_j = \overline{o_j}$ preserves bond reflection positivity without requiring integrability:

- *Single-ion anisotropy.* In the Cartesian spin-1 basis, $o = D(S^z)^2 = D\,\mathrm{diag}(1,1,0)$ is real diagonal and leaves $-\boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}$ unchanged (`models/spin1_largeD`).
- *Dimerized chains.* Bond-alternating spin-$\frac{1}{2}$ and spin-1 chains with couplings $1 \pm \delta$ group into two-spin cells with the two-spin swap F under Eq. (E4) (`models/dimerized_heisenberg`, `models/spin1_dimerized`).
- *Two-leg ladders.* Grouping each two-spin rung into a four-state site ($F = I$) places the rung coupling in o_j and the leg couplings in $-h_{\mathrm{cell}} = \sum_\alpha C_\alpha \otimes C_\alpha$, covering both the Ising ladder ($C_{1,2} = Z \otimes I$, $I \otimes Z$; `models/two_leg_tfim`) and, under a checkerboard π-rotation $u = 2iS^y$ on one spin per rung, the spin-$\frac{1}{2}$ Heisenberg ladder (`models/heisenberg_ladder`).

Reflection-positive fermion models [11] and two-dimensional lattices with reflection planes [8, 9] also satisfy (H2), with X_β acting on a transverse column in 2D.

Appendix H: Numerical Checks and Certified Models

Reflection-positivity screening. Two floating-point tests screen Hamiltonians before algebraic certification:

- *Gram test.* Tests Eq. (E2) on open chains of 4 and 6 sites, $\beta \in [0.3, 3]$, reporting the least eigenvalue of $A \mapsto \mathrm{Tr}[A\theta(A)e^{-\beta H}]$ relative to the largest, minimized over $\Theta = VK$.
- *Realignment test* (`CertifiedGaps.rp_test`). Checks Eq. (E1) via the bond realignment matrix $R[-h]_{(ij),(kl)} = -h_{(ik),(jl)}$ with an optimized on-site shift.

TABLE I. Computer-certified uniform spectral gaps $\Delta_L = E_1(L) - E_0(L)$ established via the spacetime bootstrap (`models/`; L, n_0 count unit cells), proving ground-state uniqueness and $\Delta_L \geq \Delta_{\text{cert}}$ for even L in the stated range (odd n in row 2). Lean 4 (`lean/CertifiedGaps/`, checked by `lean/scripts/check_seeds.jl`) covers rows 1, 3, 4, 8, 9 (with $(1/4, 1/20)$ proved unconditionally in `lean/SeedKernel/`), $\delta = 9/10$ in row 5, $D \in \{4, 1603/500\}$ at $n_0 = 6$ in row 6, and $J_\perp = 4$ in row 7.

Model	Cell / RP form	Certified parameters	Seed (n_0, β_0)	Δ_{cert}	Range
Spin-1 Heisenberg	1 site, G_α	isotropic $(J = 1)$	$(36, 49/4) \to (2304, 784)$	$(\log 20)/784$	$L \geq 2304$
Spin-1 open [1]	1 site, caps v_β	fields $h = 3/5$	$(120, 49) \to (1920, 392)$	$(\log 10)/392$	odd $n \geq 1921$
Staggered Heisenberg	1 site, $2iS^y$	$h \in \{1, \frac{3}{4}, \frac{1}{2}, \frac{7}{20}, \frac{1}{4}\}$	$(8,\ 21/4 \ldots 25/4)$	$1.148 \ldots 0.214$	$L \geq 8$
Dimerized spin-$\frac{1}{2}$	dimer, swap F	$\delta \in \{\frac{1}{2}, \frac{3}{10}, \frac{1}{4}\}$	$(4,\ 6 \ldots 29/4)$	0.534, 0.173, 0.058	$L \geq 4$ cells
Dimerized spin-1	dimer, KK^*	$\delta \in \{\frac{9}{10}, \frac{7}{10}\}$	$(2, 7/2)$, $(3, 17/4)$[a]	0.528, 0.218	$L \geq 2$, 4 cells
Spin-1 large-D	1 site, $D(S^z)^2$	$D = 4$; $[3.2059998, 3.206]$	$(12, 29/10)$[b], $(6, 3.048)$	0.899; 0.0010	$L \geq 6$
Heisenberg ladder	rung, $u = 2iS^y$	$J_\perp \in \{4, \frac{7}{2}, 3, \frac{5}{2}, \frac{163069}{65536}\}$	$(4,\ 2 \ldots 3)$	$1.102 \ldots 0.0015$	$L \geq 4$ rungs
Ising ladder	rung, Z_1, Z_2	$g = 3$	$(4, 3/2)$	0.636	$L \geq 4$ rungs
Longitudinal Ising	1 site, $-gX - hZ$	30 pts in $[\frac{1}{4}, 2] \times [\frac{1}{20}, 1]$	$(6,\ 2 \ldots 97/4)$	$0.0763 \ldots 3.705$	$L \geq 6$

[a] Odd cell seed $n_0 = 3$ (6 and 12 spins) enabled by the positive cell-transfer factorization (E5) ($\delta = 9/10$ uses the even seed $n_0 = 2$).
[b] Lifted from $n_0 = 6$ to $n_0 = 12$ via spatial transfer squaring and Eq. (F5), with verified dense bridges at $L = 6, 8, 10$.

The screening results match Propositions 1 and 2:

- *Spin-$\frac{1}{2}$ XYZ chain.* All eight sign patterns of $(J_x, J_y, J_z) = (\pm 1, \pm .7, \pm .4)$ follow the parity rule: zero or two ferromagnetic entries give $\geq -10^{-12}$ after optimizing V, while the other four give -0.048; the uniform-field antiferromagnet gives -0.095 and the transverse-field Ising chain gives $\geq -10^{-12}$.
- *Spin-1, Cartesian $\Theta = K$.* The Heisenberg antiferromagnet and $-(\boldsymbol{S} \cdot \boldsymbol{S})^2$ pass; the ferromagnet (-0.39) and AKLT (-0.04) fail. Over ϑ, the test is nonnegative on $\vartheta/\pi \in [-.62, 0]$ (Proposition 2) and negative on $\vartheta/\pi \in \{-1, -.88, -.75\} \cup [.12, 1]$.
- *Spin-1, every on-site $\Theta = VK$.* Minimizing over $V \in U(3)$ with twelve restarts leaves AKLT at -0.040, ULS at -0.18, and $\vartheta = -0.75\pi$ and the ferromagnet at -0.15.

A failure excludes that on-site Θ, not every block or cut.

Certified model suite and full-spectrum enclosures. Beyond the spin-1 Heisenberg chain (Ref. [2]), the repository certifies seven additional model families (Table I). For seed rings up to 16 spin-$\frac{1}{2}$ or 12 spin-1 sites, `CertifiedGaps.VerifiedSpectra` assembles symmetry blocks (whose exact multiplicities sum to d^n) in interval arithmetic, diagonalizes them in IEEE binary64, and encloses every eigenvalue via polar-factor orthogonalization, Weyl's inequality, and a priori rounding bounds ($\gamma_k = ku/(1 - ku)$ with gradual underflow) before summing $\log Z$ and iterating Eq. (F4). Perturbing a certified seed by an operator-norm bound on $\sum_j \delta o_j$ extends point certificates to coupling intervals ($D \in [3.2059998, 3.206]$ in `models/spin1_largeD` and $|J_\perp - J_*| \leq 10^{-4}$ in `models/heisenberg_ladder`), whereas a naive Weyl bound around the decoupled-rung limit $J_{\text{leg}} = 0$ loses a factor $e^{4\beta n}$ in $P_{\text{s}}(n, \beta)$ and fails at every (n, β) for $0 < J_\perp \leq 4$. Conversely, bounding the monotone factors $W_L(\beta) = e^{\beta E_0(L)} Z_L(\beta)$ over interval coverings of $(0, \infty)$ rules out all $\beta_0 > 0$ at $n_0 \in \{2, 4\}$ for the dimerized spin-$\frac{1}{2}$ chain at $\delta \leq 1/5$ and the Heisenberg ladder at $J_\perp \in \{3/2, 2\}$. At $(g, h) = (1/4, 1/20)$ in the longitudinal-field Ising chain the seed itself is proved in Lean, so that gap theorem has no hypotheses (see below).

Exact rational chains. The proof of Theorem 2 runs Eq. (F4) only until both defects are small, and a computer can check those finitely many steps in exact rational arithmetic. Write $z(a, b) = 1 - (1 - b)(1 - a)^2$, so the recurrence reads $\varepsilon_{\text{s},j+1} = f(z(\varepsilon_{\text{s},j}, \varepsilon_{\tau,j}))$ and $\varepsilon_{\tau,j+1} = f(z(\varepsilon_{\tau,j}, \varepsilon_{\text{s},j+1}))$. A *chain certificate* consists of rationals $s_0, \ldots, s_J$, $t_0, \ldots, t_J$, a rate $r > 0$, and $\varepsilon \leq 1/100$ such that

- $z_j = z(s_j, t_j) < 1$, $f(z_j) \leq s_{j+1}$, $z(t_j, s_{j+1}) < 1$ and $f(z(t_j, s_{j+1})) \leq t_{j+1}$ for $j < J$;
- $r^{2^j} z_j < 1 - z_j$ for $j < J$;
- $s_J, t_J \leq \varepsilon$ and $r^{2^J}\, 5\varepsilon \leq 1$.

If s_0 and t_0 bound the seed defects at (n_0, β_0), then s_j and t_j bound the defects at $(2^j n_0, 2^j \beta_0)$. Every even L with $2^j n_0 \leq L \leq 2^{j+1} n_0$ has $P_\tau(L, 2^j \beta_0) \geq 1 - z_j$, hence $e^{-2^j \beta_0 \Delta_L} \leq z_j/(1 - z_j) < r^{-2^j}$. The second condition therefore gives $\Delta_L > (\log r)/\beta_0$ at the first J scales, and the third hands the remaining scales to the symmetric argument with $\varepsilon_0 = \varepsilon$. The conclusion is a uniform gap $\Delta_L > (\log r)/\beta_0$ for every even $L \geq n_0$. The Julia side (`CertifiedGaps.Recurrence.rational_gap_chain`) produces such chains from certified seed defects. Lean checks them with `decide +kernel` (`lean/CertifiedGaps/ChainCheck.lean`) and proves the implication (`gapGT_of_chainCertificate` in `lean/CertifiedGaps/Chain.lean`). For the Ising seed below, $J = 8$ and $s_8 \approx 1.2 \cdot 10^{-1058}$.

An unconditional gap: the longitudinal Ising seed in Lean. For $H_L = -\sum_j Z_j Z_{j+1} - \frac{1}{4}\sum_j X_j - \frac{1}{20}\sum_j Z_j$, the Lean library `lean/SeedKernel/Ising/` proves (as `SeedKernel.Ising.ising_gap`), with no hypotheses, no `sorry` and only the standard axioms, that every even ring of $L \geq 6$ sites has a unique ground state and

$$\Delta_L > \frac{\log 13480}{97/4} \approx 0.3921. \tag{H1}$$

The seed is $(n_0, \beta_0) = (6, 97/4)$ with $1 - P_s(6, 97/4) \leq 9064 \cdot 10^{-9}$ and $1 - P_\tau(12, 97/4) \leq 3141 \cdot 10^{-8}$. These two bounds feed the chain certificate above. They are proved from the spectra of the rings of $L = 6$ and $L = 12$ sites by the following four steps.

1. *Two eigenvalues from residuals.* Two trial vectors per ring, with integer entries at scale 2^{52}, are checked by `decide +kernel`: their squared norms equal 2^{104} to relative accuracy 10^{-12}, and their residuals $\|Hx - \theta x\|$ are small enough for the next bound. For a Hermitian H with eigenvalues λ_i, diagonalizing H unitarily gives some i with $(\lambda_i - \theta)^2\|x\|^2 \leq \|Hx - \theta x\|^2$. Each check thus places an eigenvalue within $r = 10^{-10}$ of θ, at $\theta_0 \approx -6.3918$, $\theta_1 \approx -5.7966$ for $L = 6$ and at $\theta_0 \approx -12.7835$, $\theta_1 \approx -11.5932$ for $L = 12$. The two eigenvalues are distinct because $\theta_1 - \theta_0 > 2r$.

2. *All other eigenvalues by comparison.* In the Z basis, $(Hx)(\sigma) = E(\sigma)x(\sigma) - g\sum_j x(\sigma^{(j)})$, where E is the classical energy and $\sigma^{(j)}$ flips spin j. Hence $\mathrm{Re}\langle x, Hx\rangle \geq \sum_\sigma (E(\sigma) - gL)|x(\sigma)|^2$. Let B be the set of configurations with $E(\sigma) - gL < \rho$. Every x vanishing on B has $\mathrm{Re}\langle x, Hx\rangle \geq \rho\|x\|^2$. If more than $|B|$ eigenvalues were below ρ, their span would meet this subspace of codimension $|B|$ in a nonzero vector. So at most $|B|$ eigenvalues lie below ρ. A kernel count gives $|B| = 2$ at $\rho = -3.7$ ($L = 6$) and $\rho = -11.5$ ($L = 12$), both above $\theta_1 + r$. Every eigenvalue other than the two from step 1 is therefore at least ρ.

3. *Partition-function bounds.* With θ_0, θ_1, ρ as above and $\beta > 0$,

$$\begin{aligned} Z_L(\beta) &\geq e^{-\beta(\theta_0 + r)}, \\ Z_L(\beta) &\leq e^{-\beta(\theta_0 - r)}\big[1 + e^{-\beta(\theta_1 - \theta_0)} \\ &\qquad + 2^L e^{-\beta(\rho - \theta_0 + r)}\big]. \end{aligned}$$

   At $\beta = 97/4$ each correction term is at most a power of e^{-14}. Lean bounds these powers using $e > 2.7182818283$ (Mathlib's `Real.exp_one_gt_d9`).

4. *Purities.* Write the upper bound on the denominator Z_1 ($Z_6(\beta_0)$ or $Z_{12}(\beta_0)$) as $A(1 + t)$ with $A = e^{-\beta_0(\theta_0 - r)}$, and the lower bound on the numerator Z_2 ($Z_{12}(\beta_0)$ or $Z_{12}(2\beta_0)$) as $A^2 e^{-x}$. Then $Z_2/Z_1^2 \geq (1 - x)/(1 + t)^2$, and the right side is checked against both seed bounds in exact rational arithmetic.

The arithmetic is done on bit-packed integer vectors, with the $L = 12$ sums split into 64 chunks of 64 configurations. The checker's integer sums are identified with the Lean Hamiltonian (`Ising/Bridge.lean`, `Ising/Hamiltonian.lean`). No step uses `native_decide`. This point is the cheapest seed in Table I because only two states lie below ρ, whereas the seeds in the other seven model families require resolving 195–2465 low-lying states (`models/lean_seeds/`). The remaining 29 points of row 9 keep their seeds as hypotheses in `lean/CertifiedGaps/Models/IsingLongitudinal.lean`.

[1] Companion note, *A diagrammatic proof of the Haldane gap via spacetime tensor networks*, `notes/thermal-mpo-gap-note.tex` (2026).

[2] Companion note, *How the initial estimate is certified*, `notes/thermal-mpo-gap-certificate.tex` (2026).

[3] K. Osterwalder and R. Schrader, Axioms for Euclidean Green's functions, Commun. Math. Phys. **31**, 83 (1973).

[4] M. Lüscher, Construction of a selfadjoint, strictly positive transfer matrix for Euclidean lattice gauge theories, Commun. Math. Phys. **54**, 283 (1977).

[5] J. Fröhlich, B. Simon, and T. Spencer, Infrared bounds, phase transitions and continuous symmetry breaking, Commun. Math. Phys. **50**, 79 (1976).

[6] A. Jaffe and B. Janssens, Characterization of reflection positivity: Majoranas and spins, Commun. Math. Phys. **346**, 1021 (2016).

[7] F. J. Dyson, E. H. Lieb, and B. Simon, Phase transitions in quantum spin systems with isotropic and nonisotropic interactions, J. Stat. Phys. **18**, 335 (1978).

[8] J. Fröhlich, R. Israel, E. H. Lieb, and B. Simon, Phase transitions and reflection positivity. I., Commun. Math. Phys. **62**, 1 (1978).

[9] M. Biskup, *Reflection positivity and phase transitions in lattice spin models*, in *Methods of Contemporary Mathematical Statistical Physics*, Lecture Notes in Math. **1970** (Springer, 2009).

[10] E. R. Speer, Failure of reflection positivity in the quantum Heisenberg ferromagnet, Lett. Math. Phys. **10**, 41 (1985).

[11] A. Jaffe and F. L. Pedrocchi, Reflection positivity for Majoranas, Ann. Henri Poincaré **16**, 189 (2015).

[12] C. Wu and S.-C. Zhang, Sufficient condition for absence of the sign problem in the fermionic quantum Monte Carlo algorithm, Phys. Rev. B **71**, 155115 (2005).

[13] Z.-C. Wei, C. Wu, Y. Li, S. Zhang, and T. Xiang, Majorana positivity and the fermion sign problem of quantum Monte Carlo simulations, Phys. Rev. Lett. **116**, 250601 (2016).

[14] Z.-X. Li, Y.-F. Jiang, and H. Yao, Majorana-time-reversal symmetries: a fundamental principle for sign-problem-free quantum Monte Carlo simulations, Phys. Rev. Lett. **117**, 267002 (2016).